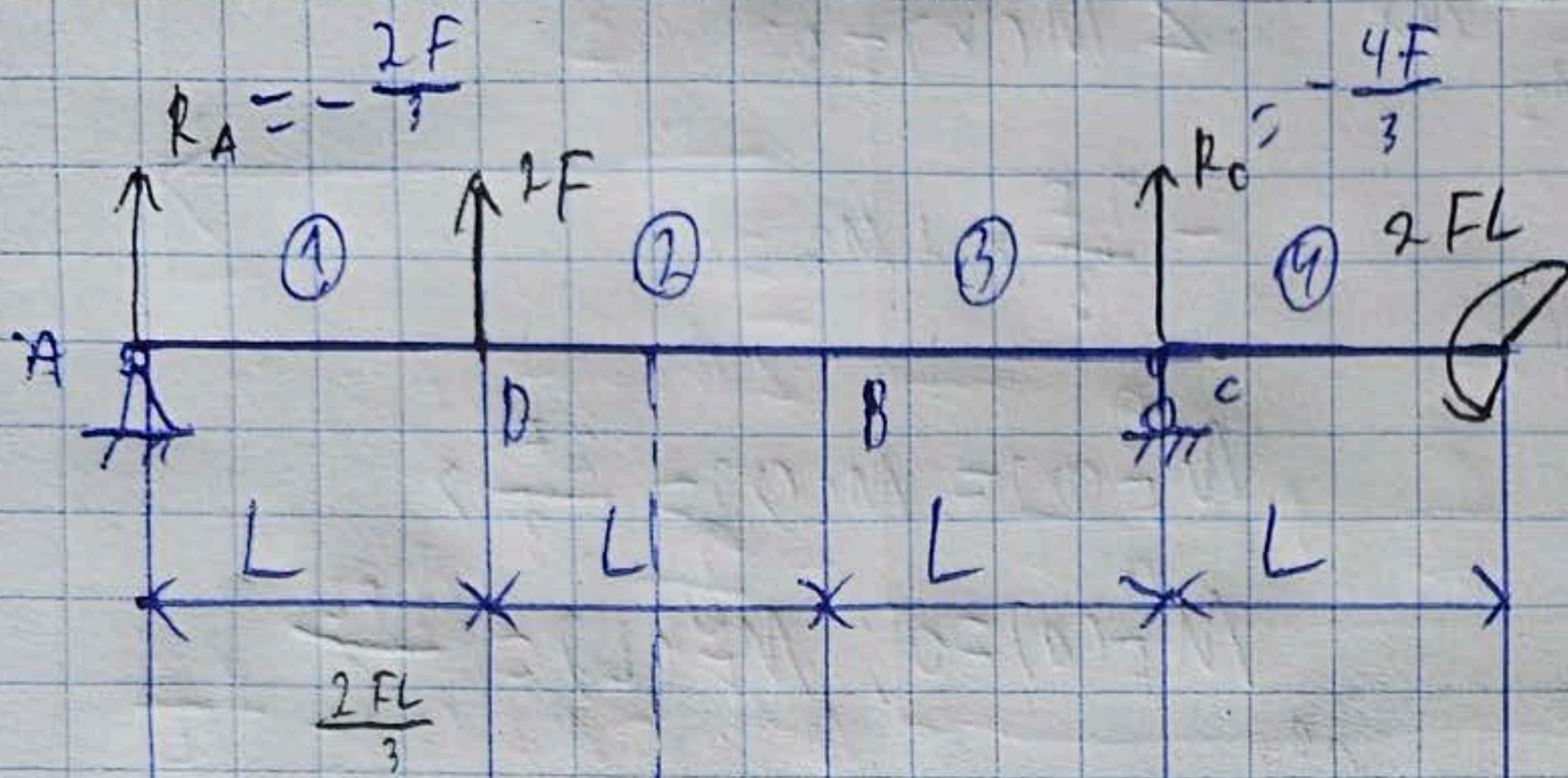


01.10

УСТЕНКО
ВАСИЛИЙ
ИДБ-24-11

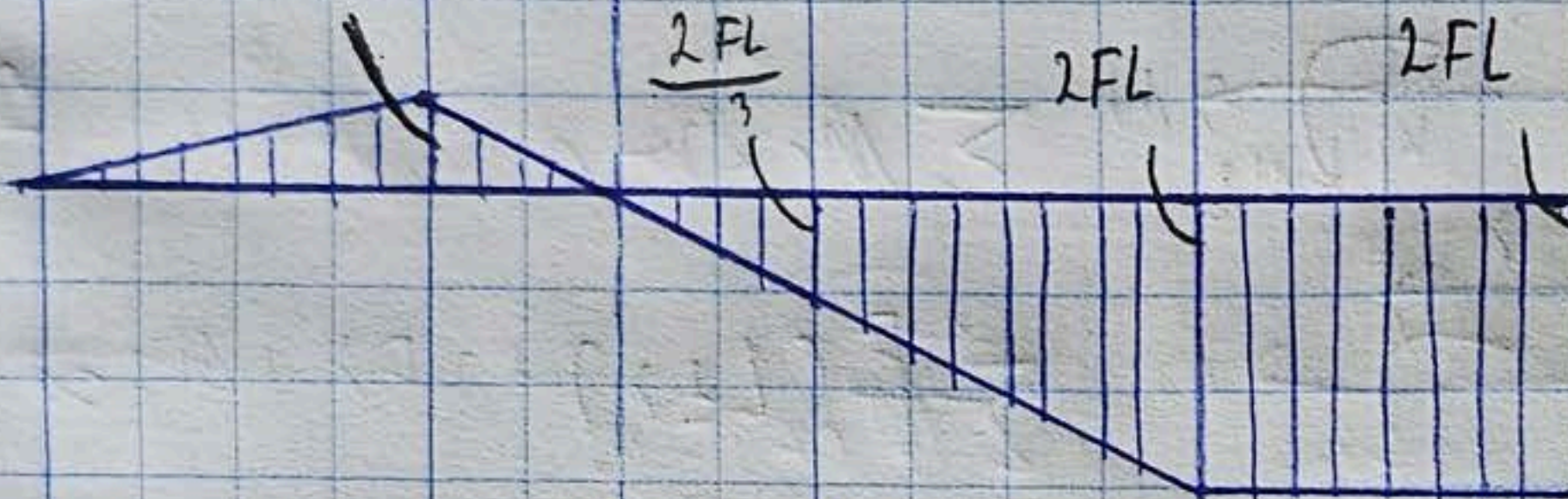
①



$$V_B = ?$$

$$\Theta_D = ?$$

M_F



$$\sum y = 0:$$

$$R_A + 2F + R_C = 0$$

$$R_A = -2F - R_C$$

$$\sum M_{(A)} = 0:$$

$$-2FL - 3LR_C - 2FL = 0$$

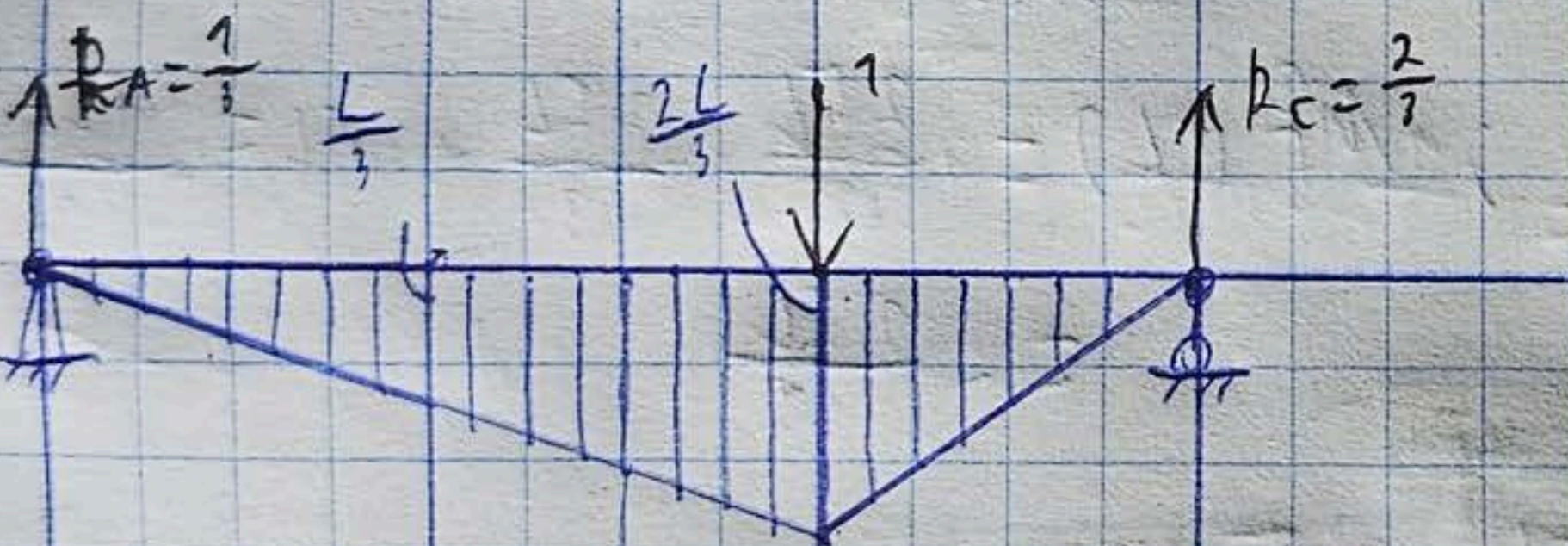
$$-3LR_C = 4FL$$

$$R_C = -\frac{4F}{3}$$

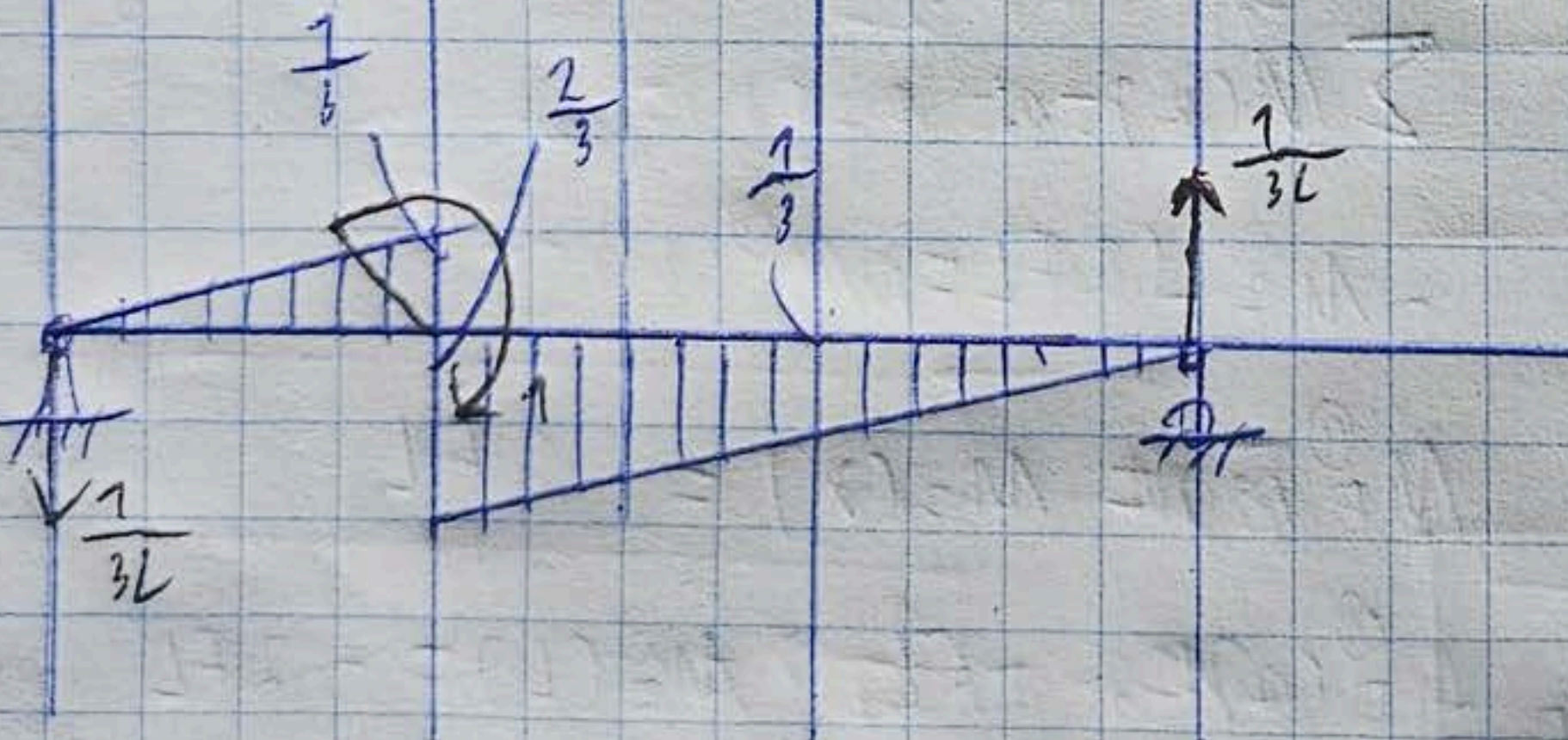
$$R_A = -2F + \frac{4F}{3} =$$

$$= -\frac{2F}{3}$$

M_1



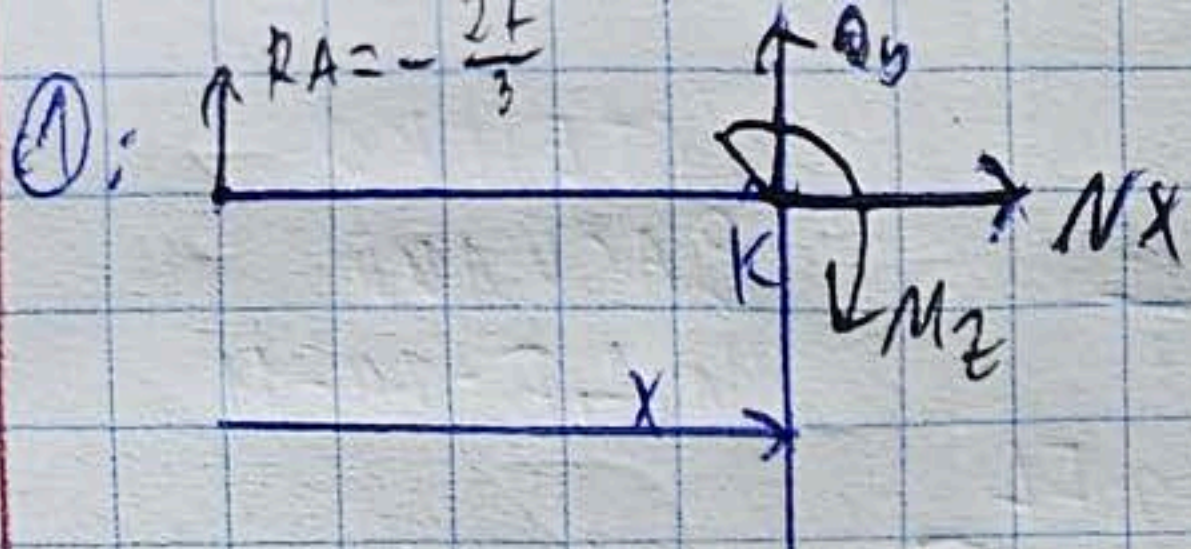
M_2



Answer: $V_B = \frac{FL^3}{2EI_z}$, $\Theta_D = \frac{2FL^2}{9EI_z}$, no absolute compliance

$\Theta_D = \frac{2FL^2}{9EI_z}$, no absolute compliance

②



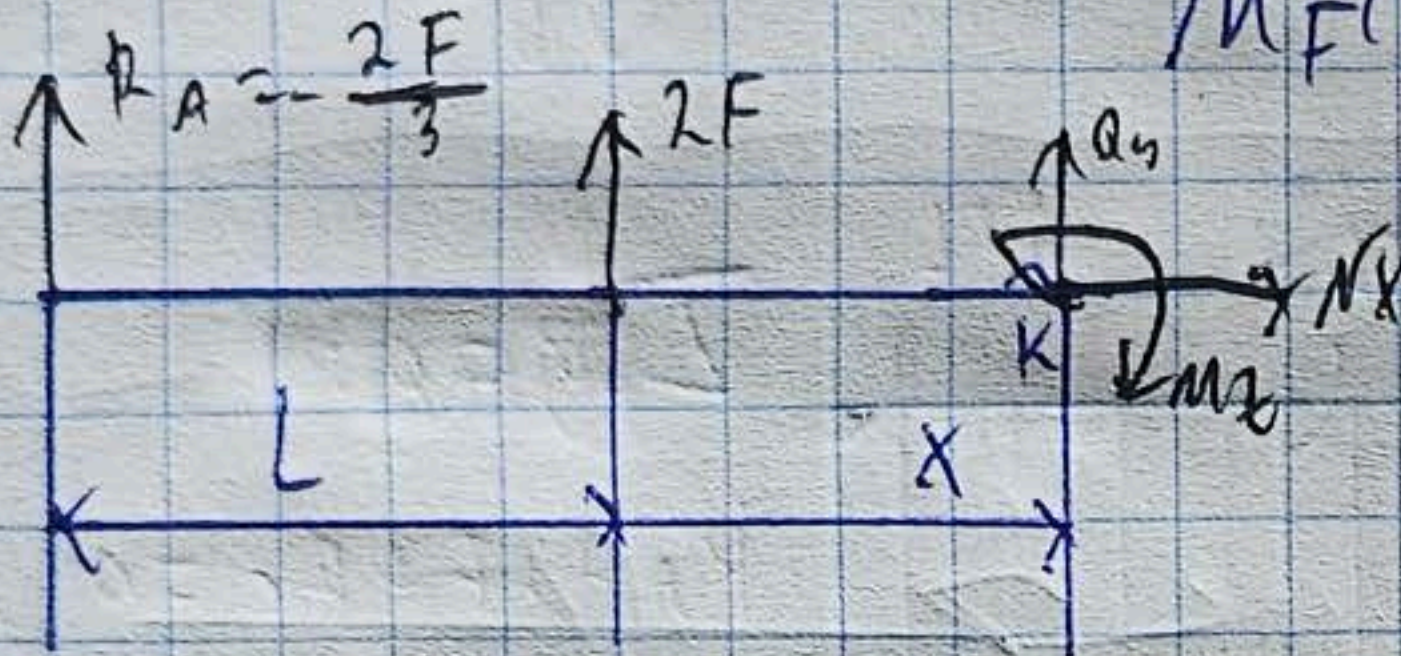
$$\sum M(x) = 0:$$

$$-\frac{2F}{3}x + M_z = 0$$

$$M_F^{(1)}(x) = M_z(x) = \frac{2F}{3}x$$

$$M_F^{(1)}(0) = 0; M_F^{(1)}(L) = \frac{2FL}{3}$$

②:



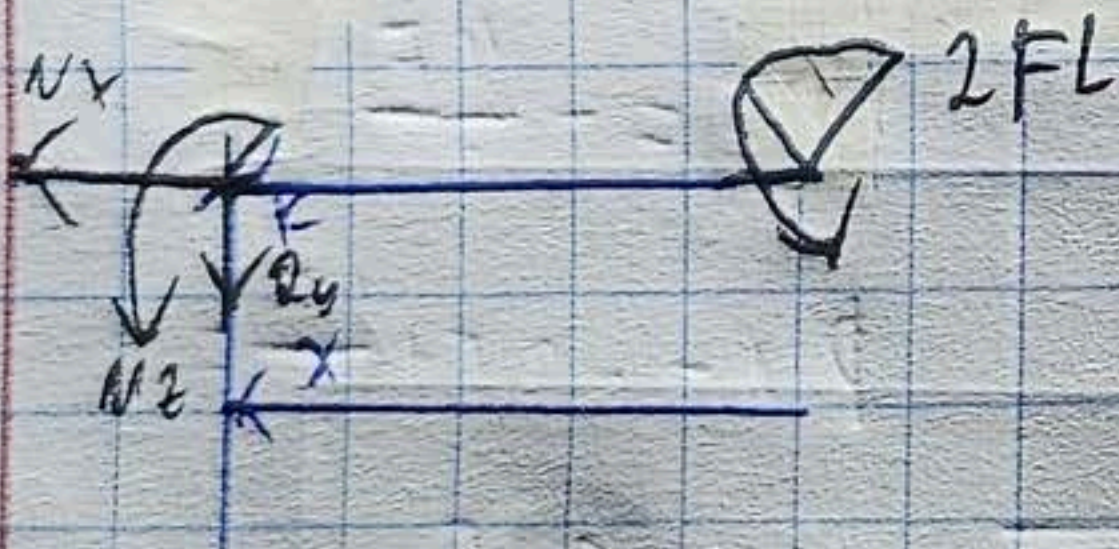
$$\sum M(x) = 0$$

$$-\frac{2F}{3}(L+x) + 2Fx + M_z = 0$$

$$M_F^{(2)}(x) = M_z(x) = \frac{2F}{3}(L+x) - 2Fx$$

$$M_F^{(2)}(0) = \frac{2FL}{3}; M_F^{(2)}(L) = \frac{2FL}{3} + \frac{2FL}{3} - 2FL = \frac{4FL - 6FL}{3} = -\frac{2FL}{3}$$

④:



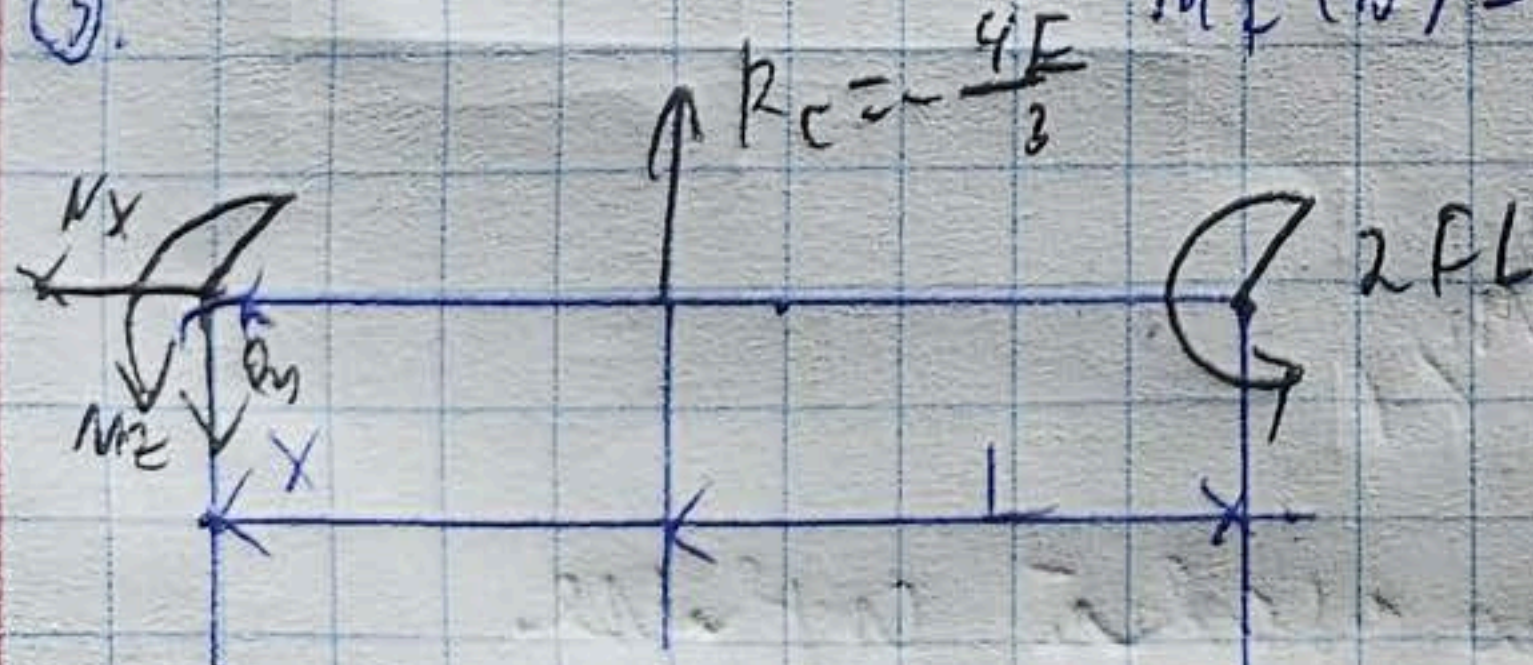
$$\sum M(x) = 0:$$

$$-M_z - 2FL = 0$$

$$M_F^{(4)}(x) = M_z(x) = -2FL$$

$$M_F^{(4)}(0) = -2FL; M_F^{(4)}(L) = -2FL$$

③:



$$\sum M(x) = 0:$$

$$-M_z + \frac{4F}{3}x - 2FL = 0$$

$$M_F^{(3)}(x) = M_z(x) = -2FL + \frac{4F}{3}x; M_F^{(3)}(0) = -2FL; M_F^{(3)}(L) = 2FL - \frac{4FL}{3} = -\frac{2FL}{3}$$

$$\sum F = 0:$$

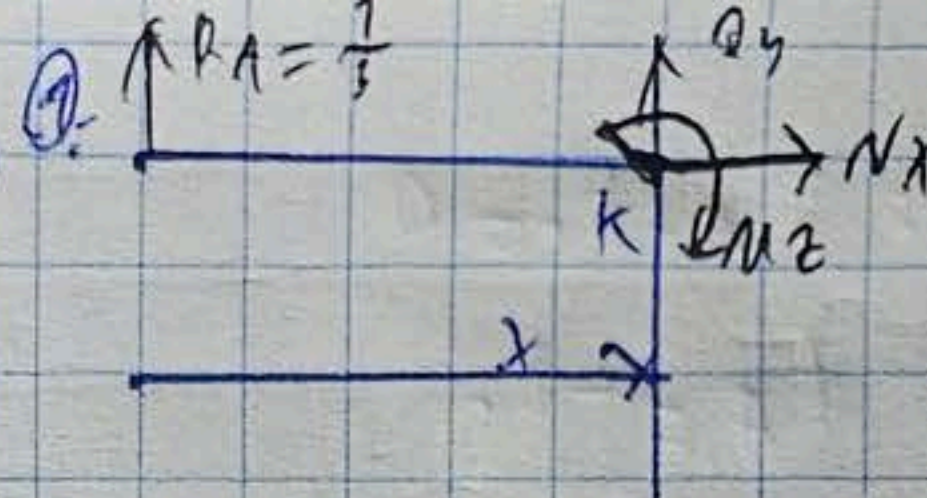
$$R_A + R_C = 1$$

$$R_A = 1 - R_C$$

$$\sum M(A) = 0:$$

$$2L - R_C L = 0$$

$$R_C = \frac{2}{3}; R_A = \frac{1}{3}$$

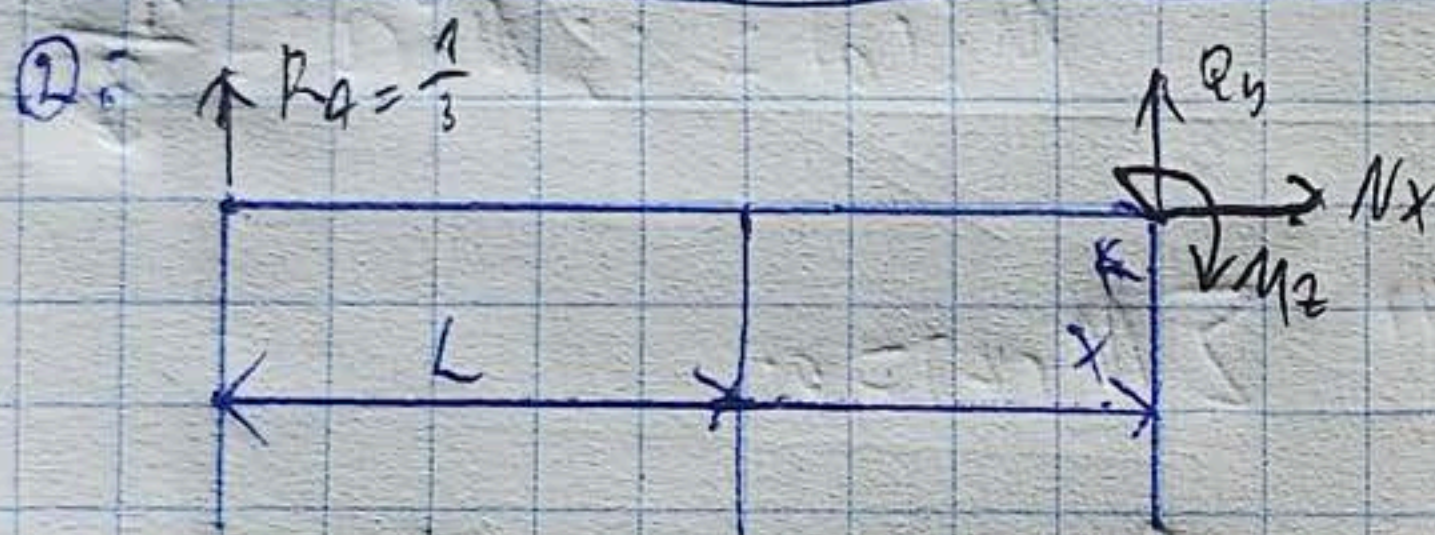


$$\sum M(x) = 0:$$

$$\frac{1}{3} - M_x = 0$$

$$M_1^0(x) = M_2(x) = -\frac{1}{3}x$$

$$M_1^0(0) = 0; M_1^0(L) = -\frac{1}{3}$$

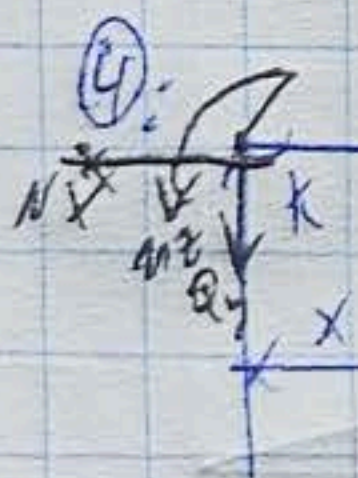


$$\sum M(x) = 0:$$

$$\frac{1}{3}(L+x) - M_x = 0$$

$$M_1^0(x) = M_2(x) = -\frac{1}{3}(L+x)$$

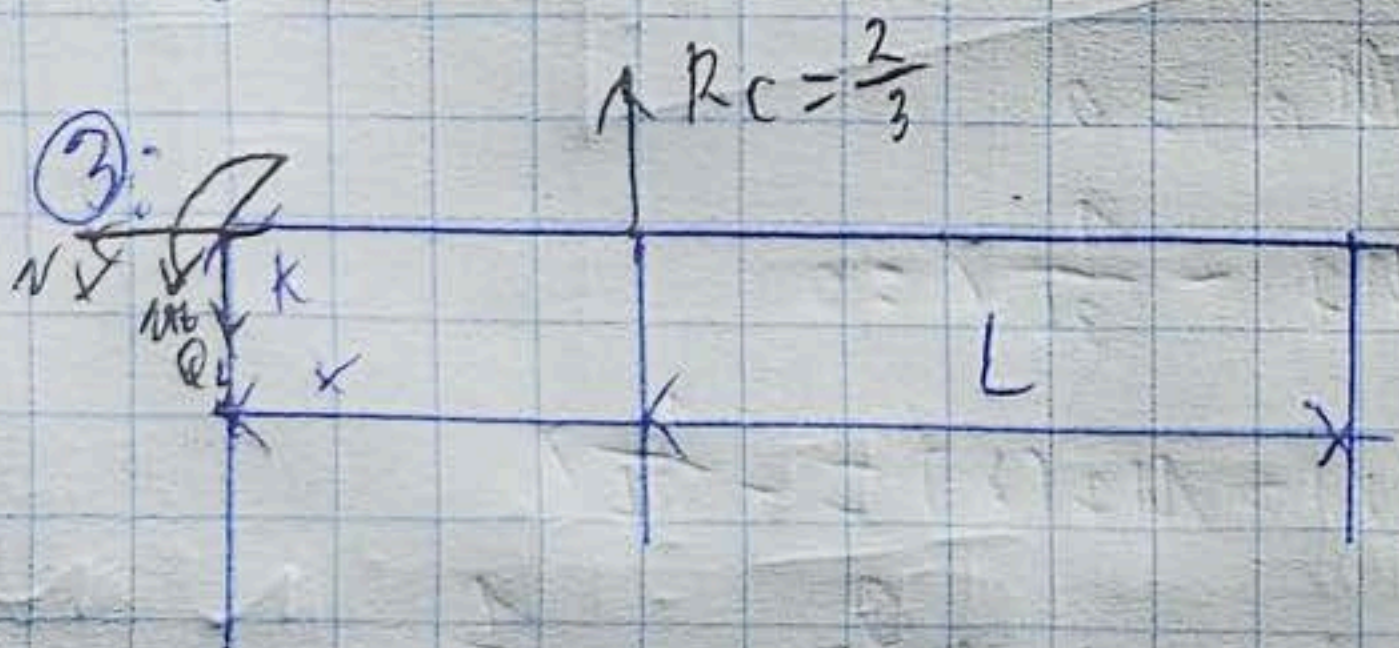
$$M_1^0(0) = -\frac{1}{3}; M_1^0(L) = -\frac{2L}{3}$$



$$\sum M(x) = 0: -M_x = 0$$

$$M_1^0(x) = M_2 = 0$$

$$M_1^0(0) = 0; M_1^0(L) = 0$$



$$\sum M(x) = 0:$$

$$-M_x - \frac{2}{3}x = 0$$

$$M_1^0(x) = M_2(x) = -\frac{2}{3}x$$

$$M_1^0(0) = 0; M_1^0(L) = -\frac{2L}{3}$$

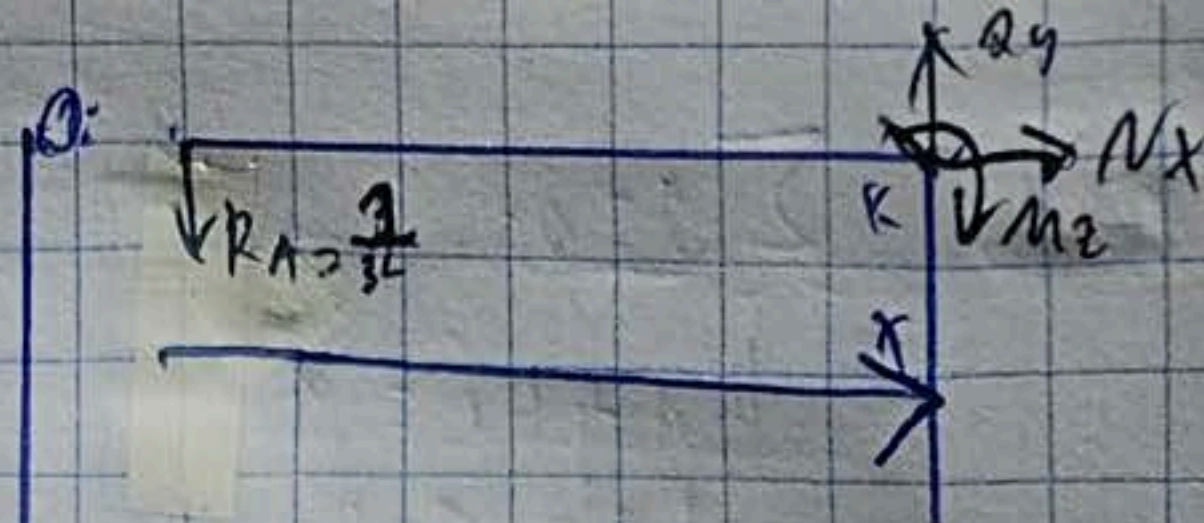
④

$$\sum \mathcal{F} = 0: R_A + R_C = 0$$

$$R_A = -R_C$$

$$\sum M_i = 0: 1 - 3LR_C = 0$$

$$R_C = \frac{1}{3L}; R_A = -\frac{1}{3L}$$



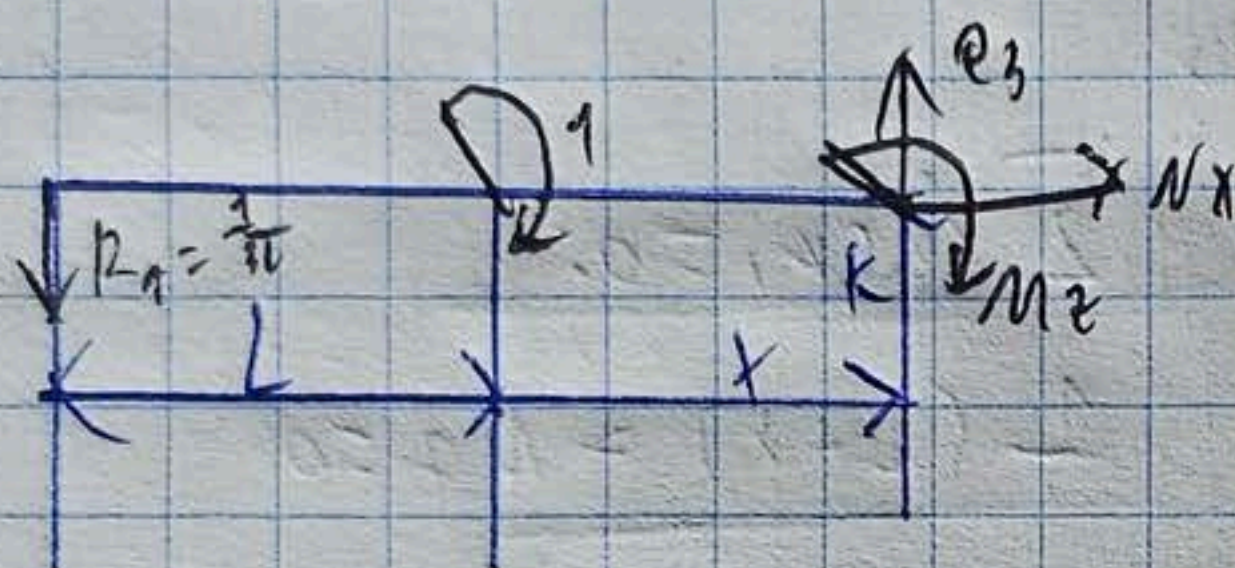
$$\sum M(x) = 0:$$

$$-\frac{1}{3L}x + M_x = 0$$

$$M_x^{\text{D}}(x) = M_x(x) = \frac{x}{3L}$$

$$M_x^{\text{D}}(0) = 0; M_x^{\text{D}}(L) = \frac{1}{3}$$

②:

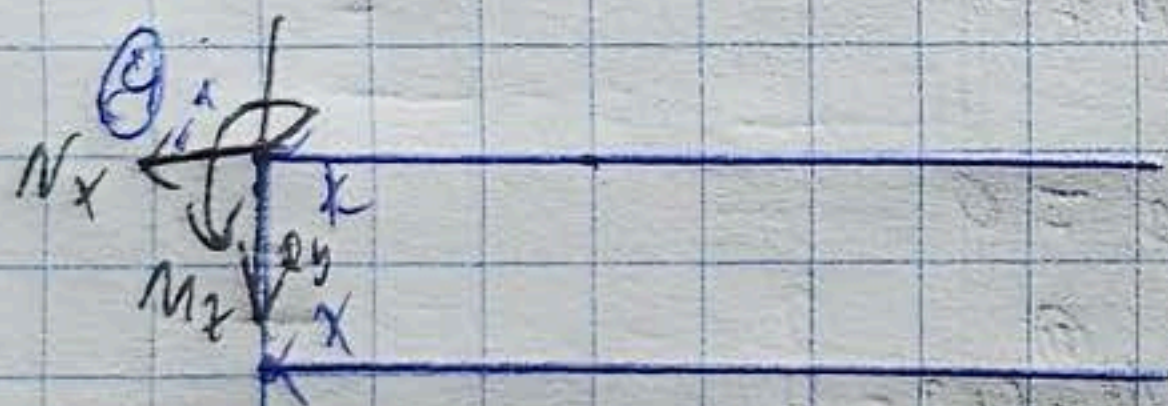


$$\sum M(x) = 0:$$

$$-\frac{1}{3L}(L+x) + 1 + M_x = 0$$

$$M_x^{\text{D}}(x) = M_x(x) = \frac{1}{3} + \frac{x}{3L} - 1 = \frac{x}{3L} - \frac{2}{3}; M_x^{\text{D}}(0) = -\frac{2}{3};$$

$$M_x^{\text{D}}(L) = -\frac{1}{3}$$

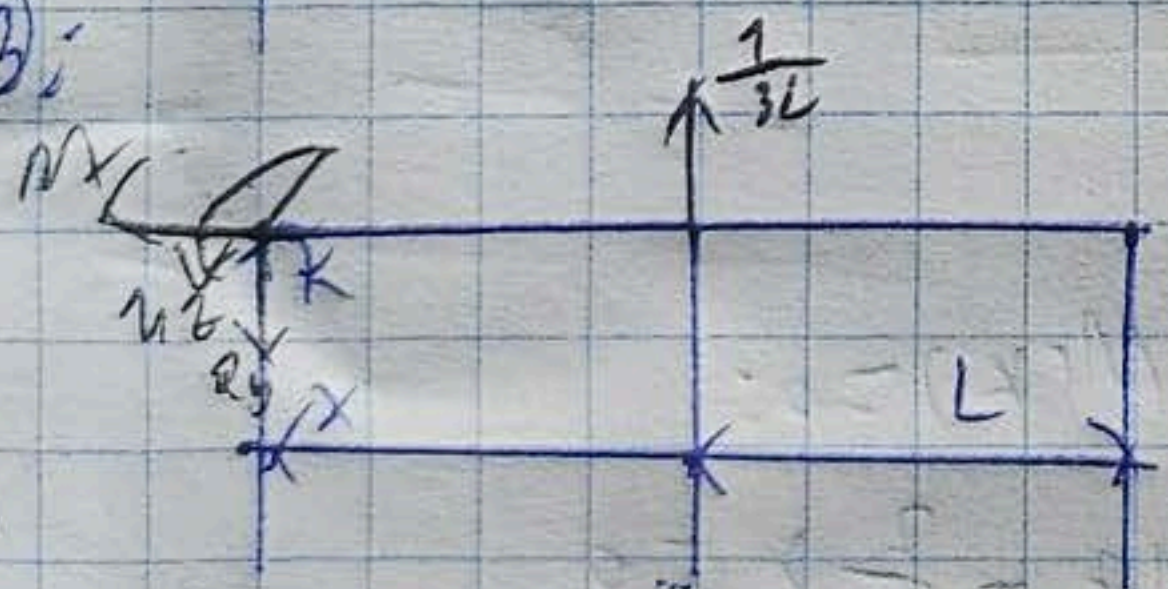


$$\sum M(x) = 0$$

$$M_x^{\text{D}}(x) = M_x(x) = 0$$

$$M_x^{\text{D}}(0) = 0; M_x^{\text{D}}(L) = 0$$

③:



$$\sum M(x) = 0$$

$$-M_x - \frac{1}{3L}x = 0$$

$$M_x^{\text{D}}(x) = M_x(x) = -\frac{1}{3L}x$$

$$M_x^{\text{D}}(0) = 0; M_x^{\text{D}}(L) = -\frac{1}{3}$$

$$A_F^{(1)} = \frac{1}{2} \cdot L \cdot \frac{2FL}{3} = \frac{FL^2}{3}; \quad \chi_C^{(1)} = \frac{2L}{3}$$

$$A_F^{(2.1)} = \frac{1}{2} \cdot \frac{L}{2} \cdot \frac{2FL}{3} = \frac{FL^2}{6}; \quad \chi_C^{(2.1)} = \frac{7L}{6}$$

$$A_F^{(2.2)} = -\frac{FL^2}{6}; \quad \chi_C^{(2.2)} = \frac{11L}{6}$$

$$A_F^{(3)} = L \cdot \left(-\frac{2}{3} - 2\right) \cdot 2FL = -\frac{4FL^2}{3}; \quad \chi_C^{(3)} = \frac{31L}{12}$$

$$A_F^{(4)} = 0$$

$$M_{1C}^{(1)} = \frac{-\frac{2L}{3}}{3} = -\frac{2L}{9}; \quad M_{7C}^{(2.1)} = \frac{-\frac{4L}{6}}{3} = -\frac{7L}{18};$$

$$M_{1C}^{(2.2)} = \frac{-\frac{11L}{6}}{3} = -\frac{11L}{18}; \quad M_{1C}^{(3)} = -\frac{2}{3} \cdot \frac{3L - 31L}{12} = -\frac{5L}{18}$$

$$M_{1C}^{(4)} = 0$$

$$v^{(1)} = \left(\frac{FL^2}{3}\right) \left(-\frac{2L}{9}\right) = -\frac{2FL^3}{27}$$

$$v^{(2.1)} = \left(\frac{FL^2}{6}\right) \left(-\frac{7L}{18}\right) = -\frac{7FL^3}{108}$$

$$v^{(2.2)} = \left(-\frac{FL^2}{6}\right) \cdot \left(-\frac{11L}{18}\right) = \frac{11FL^3}{108}$$

$$v^{(3)} = \left(-\frac{4FL^2}{3}\right) \cdot \left(-\frac{5L}{18}\right) = \frac{10FL^3}{27}$$

$$v^{(4)} = 0$$

$$v_0 = v^{(1)} + v^{(2.1)} + v^{(2.2)} + v^{(3)} + v^{(4)} \cdot \frac{1}{EI_2} = \frac{FL^3}{3EI_2}$$

many.

6

$$M_{2c}^0 = \frac{2L}{3} = \frac{2}{3}$$

$$M_{2c}^{1.1} = \frac{7}{18} - 1 = -\frac{11}{18}$$

$$M_{2c}^{1.2} = \frac{11}{18} - 1 = -\frac{7}{18}$$

$$M_{2c}^3 = \frac{31}{36} - 1 = -\frac{5}{36}$$

$$M_{2c}^4 = 0$$

$$\theta^0 = \frac{FL^2}{3} \left(\frac{2}{3} \right) = \frac{2FL^2}{27}$$

$$\theta^{1.1} = \frac{FL^2}{6} \left(-\frac{11}{18} \right) = -\frac{11FL^2}{108}$$

$$\theta^{1.2} = -\frac{FL^2}{6} \left(-\frac{7}{18} \right) = \frac{7FL^2}{108}$$

$$\theta^3 = -\frac{4FL^2}{3} \left(-\frac{5}{36} \right) = \frac{5L^2}{27}$$

$$\theta^4 = 0$$

$$\theta_D = \theta^0 + \theta^{1.1} + \theta^{1.2} + \theta^3 + \theta^4 \cdot \frac{1}{EI_2} = \frac{2FL^2}{9 \cdot EI_2}$$

по рас. Справедливо.